t3-w04 — condition C0

Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

> The curriculum objective alone. No lesson plan, no note.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	absent
Materials	what the lesson was represented with	prompt	absent
Position	where this objective sat in the sequence	prompt	absent
Method	the shape of the teaching, as event labels	prompt	absent
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cubes and Cuboids

EXPLANATION

A **cube** is a three-dimensional shape with six equal square faces. All its edges are the same length, and each face meets four other faces at right angles. Therefore:
- Number of faces: 6
- Number of edges: 12
- Number of vertices (corners): 8
- All faces are squares of the same size.

A **cuboid** (also called a rectangular prism) also has six faces, but the faces are rectangles. Opposite faces are equal, and the edges that meet at a corner are at right angles. Its properties are:
- Number of faces: 6 (three pairs of equal rectangles)
- Number of edges: 12 (four edges of each of three different lengths)
- Number of vertices: 8
- Faces come in three different sizes: length × width, length × height, and width × height.

WORKED_EXAMPLE

Worked Example

You have a cuboid whose length is 8 cm, width is 5 cm and height is 3 cm.
- Faces: there are three pairs of opposite faces.
  * Two faces are 8 cm × 5 cm.
  * Two faces are 8 cm × 3 cm.
  * Two faces are 5 cm × 3 cm.
- Edges: each dimension appears four times, so there are four edges of 8 cm, four of 5 cm and four of 3 cm – total 12 edges.
- Vertices: the corners where the three dimensions meet – always 8.

Thus this shape has the typical cuboid properties: 6 faces, 12 edges, 8 vertices, with three different face sizes.

1. A cube has an edge length of 4 cm. List the number of faces, edges and vertices it has.
answer: Faces: 6, Edges: 12, Vertices: 8

2. A cuboid has dimensions 7 cm × 4 cm × 2 cm. How many different sizes of rectangular faces does it have?
answer: Three different sizes (7×4, 7×2, 4×2)

3. True or False: All edges of a cuboid are the same length.
answer: False

4. If a shape has 6 square faces, 12 edges, and 8 vertices, what is the shape?
answer: Cube

Objective 2

Properties of Pyramids and Cones

EXPLANATION

A **pyramid** is a solid with a polygonal base and triangular faces that meet at a single point called the apex. The key properties are:
- **Base**: any polygon (commonly a square or triangle).
- **Faces**: all triangular, each sharing the apex.
- **Edges**: the base edges plus the edges from each base vertex to the apex.
- **Vertices**: the vertices of the base plus the apex.
- **Height**: the perpendicular distance from the apex to the plane of the base.

A **cone** is similar but its base is a circle. Its properties are:
- **Base**: a circle.
- **Surface**: one curved surface that tapers to the apex.
- **Edges**: only the circular edge of the base.
- **Vertices**: just the apex (the circle has no vertices).
- **Height**: the perpendicular distance from the apex to the centre of the base.

WORKED_EXAMPLE

Worked Example – Identifying properties of a square-based pyramid

You are given a solid that has a square base of side 6 cm and an apex directly above the centre of the base.

1. **Base** – The base is a square (four equal sides, four right angles).
2. **Faces** – There are four triangular faces, each joining one side of the square to the apex.
3. **Edges** – The solid has 8 edges: 4 edges of the square base and 4 edges from each corner of the base to the apex.
4. **Vertices** – There are 5 vertices: the four corners of the square base plus the apex.
5. **Height** – The height is the perpendicular distance from the apex to the base. (If the height were given as 9 cm, that would be the value to use in calculations.)

Thus, you can list the properties of this pyramid as:
- Base: square, side = 6 cm
- Faces: 4 triangles
- Edges: 8
- Vertices: 5
- Height: (given or to be measured)

1. A pyramid has a triangular base with sides 5 cm, 5 cm and 6 cm. List the number of faces, edges and vertices of the pyramid.
answer: Faces: 4 (3 triangular sides + 1 triangular base). Edges: 6 (3 base edges + 3 edges from base vertices to apex). Vertices: 4 (3 base vertices + apex).

2. A cone has a base radius of 3 cm and a height of 8 cm. Identify its base shape, number of edges, and number of vertices.
answer: Base shape: circle. Edges: 1 (the circular edge). Vertices: 1 (the apex).

3. Consider a square-based pyramid with a base side of 10 cm and a height of 12 cm. How many edges does the solid have?
answer: 8 edges (4 around the base + 4 from each base corner to the apex).

4. A cone has a base diameter of 10 cm. How many vertices does the cone have?
answer: 1 vertex (the apex).

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	grounded
Explanation the principle, rule or relationship to review	2	nothing in the input	unsourced
Worked example a problem with its solution worked through	2	nothing in the input	unsourced

Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	8	nothing in the input	**unsourced**
Answer what the learner checks against	8	nothing in the input	**unsourced**